Mohammad Mohammad

Ottawa, ON | 613-400-5719 | mohammadmohammad3@cmail.carleton.ca | linkedin.com/in/mohammadm152 | github.com/HamoodiM

EDUCATION

Carleton University, Ottawa, ON *Expected April 2028*

B.C.S. Honours in Computer Science, AI & Machine Learning Stream | GPA: 3.92/4.0

- Dean's Honours List; coursework: Computer Graphics, Data Structures & Algorithms, Operating Systems

PROJECTS

LinguaSpy: Language-Learning Spy Game *Sept 2026 – Oct 2026*

Unreal Engine 5 (C++), Python, Blender | Personal project

- Built a UE5 spy game in ~14K lines of C++ where the player keeps cover by answering locals in 4 languages over 6 missions.
- Implemented the learning core in C++: a sentence grader that tolerates accents and punctuation, and a Leitner spaced-repetition scheduler (10 min to 21 days) whose box picks the exercise type, from word tiles to free typing.
- Added a camera occlusion fade that sphere-sweeps from camera to player each tick and dithers out walls in the way.
- Versioned the save format with stepwise migrations (now v3). Old files were loading as current because UE skips writing values equal to the class default; the version now defaults to 0 and is stamped on every save.
- Dialogue and missions are UPrimaryDataAsset graphs; a custom Slate editor tab edits and validates dialogue with undo. 19 headless UE automation tests cover the grader, scheduler, saves and missions.
- Generated 7 characters on the UE mannequin skeleton with a Python Blender script, and fixed a 100x root-bone scale on FBX import by adding a real root bone and exporting in centimetres.

Monte Carlo Path Tracer *Oct 2025*

C++17, GLM, GLFW/OpenGL | Personal project, built from the Ray Tracing in One Weekend series

- Implemented ray-sphere, ray-plane and Möller-Trumbore ray-triangle intersection and a thin-lens depth-of-field camera.
- Path traces Lambertian, metal, glass (Snell, Schlick) and emissive materials with a shadow ray to each light per hit.
- Loads JSON scenes into a GLFW viewer that accumulates samples and resets when the WASD/mouse camera moves.

TitanInfer: Neural-Network Inference Engine *Feb 2026 – Mar 2026*

C++20, AVX2, CMake, GoogleTest, GitHub Actions | Personal project

- Wrote a C++20 inference library (no BLAS) with 32-byte-aligned RAII tensors, plus a standalone cache-blocked AVX2/FMA matmul kernel with its own tests and benchmark.
- Added INT8 quantization, fused Dense+activation layers, a versioned binary model format and a C API over a mutex-guarded model handle. 346 GoogleTest cases run in CI on Linux and Windows.

EXPERIENCE

QA Automation Developer *May 2026 – Present*

Government of Canada, Department of National Defence | Ottawa, ON

- Designed and wrote QASE, a C# test-automation framework that converts recorded user interactions into Excel-driven Selenium scripts, so testers no longer write each case by hand.
- Drafted user stories and specs for dashboarding, test tagging and CI/CD integration, plus a debug mode that pauses a failed run, lets the tester edit the Excel steps live and resumes from the last passing step.

IP Network Engineering Intern *Jan 2026 – Apr 2026*

Nokia | Ottawa, ON

- Worked with firmware engineers to diagnose data-path performance issues under load with packet-level analysis.
- Wrote Python and Tcl regression suites that test protocol state machines and security logic.
- Built a Python testbed-mapping tool that automated environment setup, so regression runs start from the same state.

TECHNICAL SKILLS

Languages: C++ (C++17/20), Python, C#

Engines & Graphics: Unreal Engine 5 (C++, Slate, Automation tests), Blender Python API, GLM, GLFW/OpenGL, 3D vector math

Tools: Git, CMake, Visual Studio, GoogleTest, GitHub Actions, Docker, Linux